CI Audio Enhancement Engine

Music pre-processing upstream of the implant stream - designed for how CIs represent sound. The browser app is an evaluation demo; production would integrate into a manufacturer's native mobile app.

CI users report strong speech outcomes but poor music enjoyment. Phone EQ and companion apps optimize speech-in-noise - not upstream music mastering for envelope-based, channel-limited hearing. This engine provides purpose-built DSP, MAP-aware personalization, and research tooling. The value is the processing approach - not the web browser itself.

THE OPPORTUNITY

- Unmet need: Music via CIs often sounds weak, muddy, and rhythmically flat.
- Root cause: ~16-22 envelope channels - detail is lost before phone EQ helps.
- Positioning: Upstream of Bluetooth streaming; complements clinical MAP fitting.
- Privacy: All processing on-device; no audio uploaded.

WHAT THE DEMO SHOWS TODAY

- MAP-aware shaping - 16-channel profiles, JSON import, dead-region redistribution.
- Multi-band compression - low / mid / high dynamics for CI range.
- Harmonic bass excitation - overtone correlates for envelope coding.
- High-frequency transposition - treble relocated to mid bands.
- CI Auto-Tune - 625 combinations scored via 16-channel vocoder surrogate (incl. transpose mix).
- Built-in demos - DSP Check plus Bass Focus, Melody / Harmony, and Full Mix music probes.
- Playlist and session JSON - multi-track queue; full settings export/import.
- Speech/Music modes and stereo width - one-click presets; mono collapse default.
- Mobile PWA - collapsible panels, single-column layout, installable manifest.
- Presets and A/B - Classical, Rock, Jazz; Original / Enhanced; WAV and optional vocoder export.

DEMO TODAY / PRODUCT PATH

- Today: Browser web app for demo, studies, and open review (Web Audio API).
- Production: Native iOS/Android module in the CI companion app.
- Where it runs: On the phone, upstream of wireless stream to processor.
- What transfers: Algorithms, presets, MAP mapping, auto-tune - not the browser.

ENHANCEMENT PATH

- [High] Audiogram personalisation - per-channel gain, compression, clarity.
- [High] Per-band vocoder transposition - cleaner than ring mod on tonal music.
- [High] TFS encoder - AM sidebands at the fundamental for pitch cues.
- [Med] Live mic input - delivered: concerts, TV, conversation through the chain.
- [Med] Export processed WAV - delivered: pre-process a personal library.
- [Med] Session snapshot JSON - delivered: full settings export/import.
- [Med] Optimize on loop region - auto-tune selected passage only.
- [Med] Match loudness (A/B) - fair Original / Enhanced comparison.
- [Low] Preset diff view - compare Speech vs Music sliders.
- [Low] Offline/live parity test - automated pipeline validation.

WHY THIS IS NOT "JUST EQ"

Capability	Phone EQ	CI companion app	This approach
Harmonic regeneration	-	-	**Yes**
Frequency transposition	-	-	**Yes**
MAP / electrode-aware shaping	-	Partial	**Yes**
Vocoder-in-the-loop optimization	-	-	**Yes**
Music-specific mastering chain	Generic	Speech-first	**Yes**

SUGGESTED NEXT STEPS

- Live demo - browser reference app; music demo battery; Original / Enhanced A/B with Match loudness.
- Listener study - stream processed audio to implant users vs baseline.
- Partnership - port DSP to native apps; MAP import; clinical validation.

Status and deployment

Working proof-of-concept. Vocoder is an engineering surrogate - not a clinical claim. Deliverables: reference web demo, technical paper, MAP template, DSP spec, roadmap. Browser demo for evaluation; production as native module in manufacturer iOS/Android apps.